



National Institute of Technology Rourkela
Department of Computer Science Engineering
Machine Learning (CS6225)
Mid Semester Examination, M.Tech 1st year

Duration: 2 hrs

Autumn 2024, 1st Semester

Total marks: 30

Attempt all questions

1. Answer the following questions. [2 × 5 = 10]
- Given the mappings ψ, ϕ , with any input $x \in R^d$, represent through the given functions, the difference between classification (ψ) and regression (ϕ). Assume that its a 3-class classification with classes $\{*, -, +\}$?
 - Given a model $y(x) = w^T x + b$, $w \in R^m$ being the weight vector of m dimensions and b the random error. How many minimum training and test samples are required to train the validate the accuracy of the model?
 - Identify whether the following mapping is supervised or unsupervised or neither, $f : R^d \rightarrow R$. Justify with reasons.
 - Given that E is the number of equations and X is the number of unknowns, mention the necessary conditions in terms of E and X for which the system of linear equations would be inconsistent.
 - Let $A_{n \times n}$ be a square matrix. What is the condition for A^{-1} to exist? Prove/disprove that A^{-1} is unique.
2. (a) Find the solution(s) of the following non-homogeneous system of linear equations (LeQ) if it exists using the elementary row operations. [2+3=5]

$$2x + y + 3z = 10$$

$$x + y + z = 6$$

$$x + 3y + 2z = 13$$

- (b) Given a vector space $\mathbf{V}$ with vectors $\{v_1, v_2, v_3, \dots, v_m\}$, answer the following:
- Define its basis vector β . Is basis vector unique?
 - State the condition for which the vectors in $\mathbf{V}$ are linearly independent.

- Given that $m = 3$, what will be the canonical or standard basis vector of $\mathbf{V}$?

3. Verify if the following matrix A is in row-reduced echelon form or not. Find the general solutions of the equivalent LeQ of A by using the standard procedure. [5]

$$A = \begin{bmatrix} 1 & 3 & 0 & 0 & 3 \\ 0 & 0 & 1 & 0 & 9 \\ 0 & 0 & 0 & 1 & -4 \end{bmatrix}$$

4. Given a linear regression model $Y = W^T X + \beta$ with the following information.
 The initial weight vector: $W = [2, 3, 1, 5]^T$; Input training sample: $X = [5.6, 6.7, 4, 8]^T$;
 Random error: $\beta = [1.2, 1.3, 2.1, 1.5]^T$ and Training output: $Y = [y_{a1}, y_{a2}, \dots, y_{a4}]^T$.
 Assuming that the loss function as $\mathcal{L}(MSE)$ to be *minimum*, find the values of Y vector, given that the model is trained till first pass only. [MSE = Mean Square Error] [5]
5. Determine if the following matrix A is invertible or not? If 'yes', then find the inverse of A using Gaussian elimination technique. [5]

$$A = \begin{bmatrix} 1 & 0 & 2 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 2 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$